



**THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE**  
**SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING**

DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

**B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION**  
**COURSE CODE:** ICT 415/EEE 415/CPE 415 **COURSE TITLE:** System Analysis and

Modelling

**UNITS: 3**

**TIME ALLOWED: 3 Hours**

**Instruction:** Answer ALL QUESTIONS in Section A and ONE QUESTION EACH from Section B, C and D.

**SECTION A**

1. What is the primary purpose of mathematical system modeling? (A) To replace experimental testing entirely. (B) To simplify and analyze real-world systems using equations. (C) To eliminate uncertainties in engineering systems. (D) To focus only on theoretical concepts.
2. Which of the following is NOT a common tool for system modeling? (A) Differential equations. (B) Probability distributions. (C) Literary analysis. (D) Block diagrams
3. The first step in mathematical modeling is typically: (A) Solving equations. (B) Defining the problem and assumptions. (C) Validating experimental data. (D) Writing computer code
4. Which type of model describes systems using probabilities and randomness? (A) Deterministic. (B) Stochastic. (C) Linear. (D) Static
5. Validation in modeling ensures: (A) The model matches real-world behavior. (B) The model uses the most complex equations. (C) The model is visually appealing. (D) The model runs quickly on computers
6. The general form of a population growth model with birth rate  $\beta$  and death rate  $\delta$  is: (A)  $\frac{dP}{dt} = \beta - \delta$  (B)  $\frac{dP}{dt} = (\beta - \delta)$  (C)  $\frac{dP}{dt} = \beta P - \delta$  (D)  $\frac{dP}{dt} = \beta P \delta$
7. In a population model with immigration  $I$ , the ODE becomes: (A)  $\frac{dP}{dt} = (\beta - \delta)P$  (B)  $\frac{dP}{dt} = (\beta - \delta)P + I$  (C)  $\frac{dP}{dt} = I - E$  (D)  $\frac{dP}{dt} = \beta P - I$
8. The solution to  $\frac{dP}{dt} = kP$  is: (A)  $P(t) = P_0 e^{kt}$  (B)  $P(t) = P_0 e^{kt}$  (C)  $P(t) = P_0 + kt$  (D)  $P(t) = \frac{k}{P_0} t$
9. If emigration  $E$  is added to the model, the ODE becomes: (A)  $\frac{dP}{dt} = (\beta - \delta)P - E$  (B)  $\frac{dP}{dt} = \beta P + E$  (C)  $\frac{dP}{dt} = (\beta - \delta)P + I - E$  (D)  $\frac{dP}{dt} = E - \delta P$
10. A population model with carrying capacity  $K$  is described by: (A) Exponential growth (B) Logistic growth. (C) Geometric growth. (D) Poisson growth
11. The \_\_\_\_\_ distribution models the time between independent events with rate  $\lambda$ .
12. The number of trials until the first success is modeled by the \_\_\_\_\_ distribution.
13. The Poisson distribution's mean and variance are both equal to \_\_\_\_\_.
14. The \_\_\_\_\_ distribution describes rare events in a fixed interval.
15. A Bernoulli trial has only two outcomes: \_\_\_\_\_ and \_\_\_\_\_.
16. The \_\_\_\_\_ rule states  $P(A \cap B) = P(A|B)P(B)$ .
17. A CDF must be \_\_\_\_\_ continuous.
18. Two events are \_\_\_\_\_ if  $P(A \cup B) = P(A)P(B)$ .
19. The \_\_\_\_\_ probability of an event is its probability before observing new evidence.
20. The \_\_\_\_\_ function gives  $P(X = x)$  for a discrete random variable



21. Stochastic models account for randomness, while deterministic models do not. **TRUE/FALSE**
22. The Binomial distribution describes continuous data. **TRUE/FALSE**
23. The Geometric distribution models the number of successes in  $n$  trials. **TRUE/FALSE**
24.  $P(A|B) = P(B|A)$  for any events A and B. **TRUE/FALSE**
25. The empirical probability is not based on theoretical models. **TRUE/FALSE**

### SECTION B

1. (a). (i). Define the Poisson distribution and state its probability mass function (PMF). (2 marks)  
 (ii). Name two real-world scenarios where Poisson distribution is applicable. (2 marks)  
 (iii). What is the relationship between the Poisson parameter  $\lambda$  and its mean/variance? (2 marks)
- (b). (i). Define the CDF  $F_X(x)$  of a random variable X. What are its key properties? (3 marks)  
 (ii). How does the CDF of a discrete variable differ from that of a continuous variable? (2 marks)  
 (iii). The lifetime of a machine part has CDF  $F_T(t) = 1 - e^{-t/100}$ . What does  $F_T(50)$  represent? (3 marks)
- (c). In a smart city project, the number of connected IoT devices (e.g., sensors, cameras) grows due to new deployments (**birth rate**) but also decreases due to device failures or obsolescence (**death rate**). Let  $N(t)$  represent the number of active devices at time  $t$ . Assume the device birth rate (new deployments) is proportional to the current number of devices with constant  $\beta > 0$ , and the death rate (failures) is proportional to  $N(t)$  with constant  $\delta > 0$ .  
 (i). Derive the first-order ordinary differential equation (ODE) for  $N(t)$ . Solve the ODE using the separation of variables method, assuming an initial condition  $N(0) = N_0$ . Express the solution in terms of  $\beta$ ,  $\delta$ , and  $N_0$ .  
 (ii). Find the condition under which the number of devices grows indefinitely ( $\lim_{t \rightarrow \infty} N(t) = \infty$ ) or decays to zero. Discuss implications for network planning.  
 (iii). If  $\beta = 0.1/\text{month}$  and  $\delta = 0.03/\text{month}$ , calculate the time for the device count to double.  
 (iv). If  $N_0 = 500$ ,  $\beta = 0.08$ , and  $\delta = 0.02$ , estimate  $N(t)$  after 12 months.  
 (Solve all of question 1.c using the mathematical modelling steps discussed in class) (11 marks)
2. (a). (i). Write the PDF of a normal distribution  $N(\mu, \sigma^2)$  and explain the roles of  $\mu$  and  $\sigma$ . (3 marks)  
 (ii). Why is the normal distribution called the "bell curve"? (3 marks)  
 (iii). State the **68-95-99.7 rule** and its significance. (3 marks)
- (b). (i). What is a PMF? How does it differ from a PDF? (2 marks)  
 (ii). Give an example of a discrete random variable and its PMF. (2 marks)
- (iii). In a gene mutation study, X counts mutations per cell with PMF  $P(X = k) = \frac{e^{-2} 2^k}{k!}$ . What distribution is this? (2 marks)
- (c). In a cloud computing cluster, the number of active server nodes  $N(t)$  changes over time due to the following factors: **Birth rate** ( $I$ ):  $\rightarrow$  New nodes are added at a rate proportional to the current nodes (e.g., auto-scaling). **Death rate** ( $\delta$ ):  $\rightarrow$  Nodes fail at a rate proportional to  $N(t)$ . **Immigration** ( $I$ ):  $\rightarrow$  External nodes join the cluster at a constant rate (e.g., migrated workloads), and **Emigration** ( $E$ ):  $\rightarrow$  Nodes are removed at a constant rate (e.g., maintenance).  
 (i). Derive the first-order ODE for  $N(t)$  including all four terms ( $\beta, \delta, I, E$ ).  
 (ii). Solve the ODE using separation of variables, assuming initial condition  $N(0) = N_0$ .  
 (iii). Under what condition does  $N(t)$  grow indefinitely  
 (iv). If  $\beta = 0.05/\text{hr}$ ,  $\delta = 0.02/\text{hr}$ ,  $I = 10\text{nodes/hr}$ ,  $E = 5\text{nodes/hr}$ , and  $N_0 = 100$ : Calculate  $N(t)$  after 6 hours  
 (Solve all of question 2.c using the mathematical modelling steps discussed in class) (10 marks)



### SECTION C

3. (a). A call center receives an average of 5 calls per minute, using the Poisson distribution,
- Calculate the probability of receiving exactly 3 calls in a minute. (3 marks)
  - Why might the Poisson assumption break down during a holiday sale? (4 marks)
- (b). (i). If  $P(A|B) = P(A)$ , what does this imply about events A and B? (2 marks)
- A class has 60% girls and 40% boys. 30% of girls and 50% of boys play chess.
- What is the probability a randomly selected student plays chess? (4 marks)
  - If a chess player is selected, what's the probability they're a girl? (2 marks)
- (c). A digital sensor transmits 50 data bits per second. Each bit has a 4% probability ( $p = 0.04$ ) of being corrupted due to noise. Assume errors occur independently. Using the Binomial distribution model, compute the following:
- Calculate the expected number of corrupted bits and variance of the corrupted bits per second. (4 marks)
  - Calculate the probability of no more than 3 corrupted bits in one second. (3 marks)
  - Determine the probability of at least 1 corrupted bit in one second. (3 marks)
4. (a). A batch of 1000 voltage regulators is tested for defectiveness. Each regulator has a 2% probability ( $p = 0.02$ ) of being defective (failure) and 98% probability ( $q = 0.98$ ) of functioning correctly (success). This is a Bernoulli trial for each unit.
- Define the Bernoulli random variable  $X$  for this scenario. (3 marks)
  - Calculate the expected number of defective regulators in the batch. (2 marks)
  - Compute the variance of defective units. (3 marks)
- (b). A smartphone manufacturer tests the durability of a new model's charging port. Each time a user plugs in the charger, there is a 5% probability ( $p = 0.05$ ) that the port will fail due to wear and tear. Failures occur independently. Use the geometric distribution model to answer the following questions:
- Calculate the probability that the port fails exactly on the 10th use. (3 marks)
  - Find the probability that the port lasts at least 20 uses before failing. (3 marks)
  - Determine the expected number of uses until failure. (2 marks)
- (c). The cumulative distribution function (CDF) of the time  $T$  (in minutes) it takes a bank teller to serve a customer is defined by:

$$F_T(t) = \begin{cases} 0 & t < 2 \\ A(t - 2) & 2 \leq t < 6 \\ 1 & t \geq 6 \end{cases}$$

- Determine the value of  $A$  that makes  $F_T(t)$  a valid CDF. (3 marks)
- Using the value of  $A$  from part (a), compute  $P(T > 4)$ . (3 marks)
- Using the same  $A$ , find  $P(3 \leq T \leq 5)$ . (3 marks)

### SECTION D

5. (a). A cloud computing company uses servers from 3 manufacturers to host its services: 50% of servers are from Company A with  $P(\text{Fail} | A) = 0.01$ , 30% from Company B with  $P(\text{Fail} | B) = 0.03$  20% from Company C with  $P(\text{Fail} | C) = 0.05$ . A server randomly fails. Answer the following questions:
- Use the Law of Total Probability to find the probability that any randomly selected server fails. (2 marks)
  - If a server fails, what is the probability it came from Company B. (3 marks)
  - Based on the results, which manufacturer should the company prioritize for replacement? Justify. (2 marks)
- (b). The number of telephone calls arriving at a switchboard during any 10-minute period follows a Poisson distribution with rate parameter  $\lambda = 2$ .
- What is the probability that more than three calls will arrive during a 10-minute period (4 marks)
  - What is the probability that no calls will arrive during a 10-minute period? (1 marks)
  - How does  $P(X > 3)$  change if  $\lambda$  increases from 2 to 3? Interpret the result. (4 marks)



- (c). The cumulative distribution function (CDF) of the lifetime (in years) of a piece of equipment is given by:

$$f_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x/5} & x \geq 0 \end{cases}$$

- (i). What is the expected lifetime of the equipment? (3 marks)  
 (ii). What is the variance of the lifetime of the equipment? (4 marks)  
 (iii). Find the probability that the equipment lasts more than 10 years. (2 marks)

6. (a). A cybersecurity system detects malware in email attachments. Historically:

- 2% of emails contain malware, i.e.,  $P(\text{Malware}) = 0.02$ .
- If an email has malware, the system flags it 95% of the time, i.e.,  $P(\text{Flagged}|\text{Malware}) = 0.95$ .
- If an email is clean, the system falsely flags it 3% of the time  $P(\text{Flagged}|\text{Clean}) = 0.03$ .

- (i). What is the probability that a flagged email actually contains malware? i.e.,  $P(\text{Malware}|\text{Flagged})$ . (3 marks)  
 (ii). Should the IT team prioritize reviewing flagged emails? Justify. (2 marks)  
 (iii). If the system's false-positive rate drops to 1%, how does this affect  $P(\text{Malware}|\text{Flagged})$ ? (3 marks)  
 (b). An insurance company wants to assess the risk of car accidents for drivers aged 18–25 in a city. They use two approaches:  
**Empirical Probability:** Use past accident data from 10,000 drivers in this age group (1,200 had accidents last year).  
**Axiomatic Probability:** Model accidents using a theoretical risk function  $P(A) = 1 - e^{-0.15}$  (derived from traffic laws and vehicle safety standards).  
 Use the above information to answer the following questions  
 (i). Calculate the accident probability for each method. (2 marks)  
 (ii). Compare the results. Which method is most reliable? Why? (2 marks)  
 (iii). Is there a discrepancy in the values of the empirical and axiomatic values? propose reasons for the discrepancy if there is. (2 marks)  
 (c). A student models the lifetime  $X$  (in arbitrary units) of a company's equipment using a random variable with the probability density function (PDF):

$$f_X(x) = \begin{cases} \frac{x e^{-x/10}}{100} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- (i). Show that  $f_X(x)$  is a valid PDF. (4 marks)  
 (ii). Find the probability that the equipment's lifetime exceeds 20. (3 marks)  
 (iii). Calculate the expected lifetime  $E[X]$ . (4 marks)